

Project Executable Files

Date	29 june 2026
Team ID	SWTID-2026-6540
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	YES
2	README / Setup Guide (instructions to run the project)	YES
3	requirements.txt / package.json / dependency file	NO
4	Database schema / seed files (if applicable)	NO
5	Environment configuration file (.env.example or similar)	NO
6	Deployed application URL (if hosted)	NO
7	APK / Executable binary (if applicable for mobile/desktop apps)	YES
8	Dockerfile / Containerization config (if applicable)	NO
9	Test files and test results	YES
10	Demo video or walkthrough (if required)	YES

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
OptiCrop/
|
|— app.py          # Main Flask application
|— model.pkl       # Trained Machine Learning model
|— requirements.txt # Project dependencies
|— README.md       # Project documentation
|
|— dataset/
|   └─ crop_data.csv # Crop dataset used for training
|
|— static/
|   └─ css/
|       └─ style.css # Application styling
|       └─          # Client-side functionality
|       └─
|       └─ images/
|           └─ logo.png # Project logo
|           └─ background.jpg # Background image
|
|— templates/
|   └─ index.html # Home page
|   └─ result.html # Prediction result page
|
|— notebooks/
|   └─ model_training.ipynb # Data analysis and model training
|
|— screenshots/
|   └─ home_page.png
|   └─ prediction_page.png
|   └─ result_page.png
|
└─ docs/
    └─ Project_Report.pdf
    └─ SRS_Document.pdf
    └─ Presentation.pptx
    └─ ER_Diagram.png
```

- └─ Data_Flow_Diagram.png
- └─ Architecture_Diagram.png
- └─ Testing_Report.pdf

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	
Login Credentials (Demo)	
Platform / Hosting Provider	Local hosting
Repository Link	https://github.com/bhaskarkovela/Opti-crop-predictor
Demo Video Link	https://drive.google.com/file/d/1evEUwcRgNGyisxBy7YVrj3e7tQ7Sm3kD/view?usp=sharing

Step 4: Run Instructions



OptiCrop – Instructions to Run in Local

Smart Agricultural Production Optimization Engine

1 Clone the Repository

Open terminal or command prompt and run:

```
git clone https://github.com/your-username/OptiCrop.git
cd OptiCrop
```

2 Create a Virtual Environment (Optional but Recommended)

Run the following commands:

```
python -m venv venv
```

On Windows:

```
venv\Scripts\activate
```

On macOS / Linux:

```
source venv/bin/activate
```

3 Install Required Dependencies

Run the following command:

```
pip install -r requirements.txt
```

4 Verify Project Structure

Make sure the following files/folders are present:

- ✓ app.py
- ✓ model.pkl
- ✓ templates/
- ✓ static/
- ✓ dataset/crop_data.csv

5 Run the Application

Start the Flask server by running:

```
python app.py
```

Note: Make sure you are inside the project directory (OptiCrop) while running the above command.

6 Access the Application

Open your web browser and go to:

```
http://127.0.0.1:5000/
```

i Additional Information

- ▶ The app runs on the local server at port 5000 by default.
- ▶ Make sure your system has Python 3.8 or above installed.
- ▶ If port 5000 is already in use, you can change the port in app.py:

```
app.run(debug=True, port=5001)
```
- ▶ Ensure that model.pkl is in the root directory.

⚠ Troubleshooting

Module not found errors?
Run: `pip install -r requirements.txt`

Port already in use?
Change the port in app.py (e.g., port=5001)

Model not loading?
Ensure model.pkl exists in the project root directory.

🚀 You're All Set!

The OptiCrop application should now be running successfully on your local machine.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1		
2		
3		